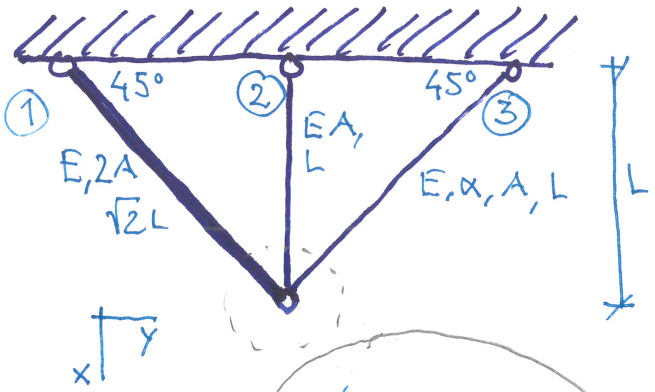
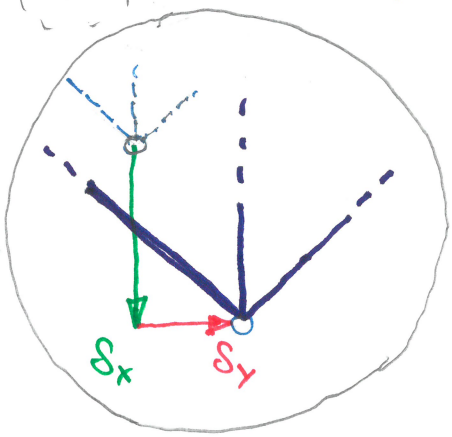
Exempel:



Bestäm knutens förskjutning om stång 3 ges en temp.höjn. $\Delta T > 0$.
Vilka stångkrafter fås då?

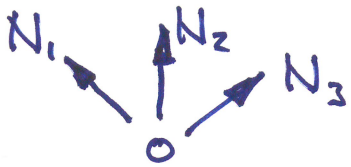
Linjärt termoelastiskt mtrl (E, α): *
Fortf. linjärt system/problem.



Ansätt knutförskjutningar δ_x, δ_y.
Identifiera stångstorheter (* nu med: $\varepsilon = \frac{\sigma}{E} + \alpha \Delta T$)

Stång	komp.		konst. samb. *	jvkt.
(i)	δ_i	$\varepsilon_i = \frac{\delta_i}{L_i}$	$\sigma_i = E_i(\varepsilon_i - \alpha_i \Delta T_i)$	$N_i = \sigma_i A_i$
①	$\frac{\delta_x+\delta_y}{\sqrt{2}}$	$\frac{\delta_x+\delta_y}{2L}$	$\frac{\delta_x+\delta_y}{2L}E$	$\frac{\delta_x+\delta_y}{2L}E\cdot 2A = N_1$
②	δ_x	$\frac{\delta_x}{L}$	$\frac{\delta_x}{L}E$	$\frac{\delta_x}{L}EA = N_2$
③	$\frac{\delta_x-\delta_y}{\sqrt{2}}$	$\frac{\delta_x-\delta_y}{2L}$	$\left(\frac{\delta_x-\delta_y}{2L} - \alpha\Delta T\right)E$	$\left(\frac{\delta_x-\delta_y}{2L} - \alpha\Delta T\right)EA = N_3$

Jämvikt för knuten:



$\uparrow: \frac{N_1}{\sqrt{2}} + N_2 + \frac{N_3}{\sqrt{2}} = 0$

$\leftarrow \frac{N_1}{\sqrt{2}} - \frac{N_3}{\sqrt{2}} = 0$

Med N_1, N_2 & N_3 (ur schemat ovan) insatta ...

...fås nu (flytta över termer med $\alpha \cdot \Delta T$ till HL)

$$\frac{EA}{L}\left(\frac{\delta_x+\delta_y}{\sqrt{2}}+\delta_x+\frac{\delta_x-\delta_y}{2\sqrt{2}}\right)=\frac{EA\alpha\Delta T}{\sqrt{2}}$$

$$\frac{EA}{L}\left(\frac{\delta_x+\delta_y}{\sqrt{2}}-\frac{\delta_x-\delta_y}{2\sqrt{2}}\right)=-\frac{EA\alpha\Delta T}{\sqrt{2}}$$

} 2 ekv. & 2 obek.

På matrisform:

$$\frac{EA}{L}\frac{1}{2\sqrt{2}}\begin{bmatrix}2\sqrt{2}+3 & 1\\ 1 & 3\end{bmatrix}\begin{bmatrix}\delta_x\\ \delta_y\end{bmatrix}=\frac{EA\alpha\Delta T}{\sqrt{2}}\begin{bmatrix}1\\ -1\end{bmatrix}$$

med lösning: $1/\det$.

$$\begin{bmatrix}\delta_x\\ \delta_y\end{bmatrix}=\frac{L}{EA}\frac{1}{3+2\sqrt{2}}\begin{bmatrix}3 & -1\\ -1 & (3+2\sqrt{2})\end{bmatrix}\begin{bmatrix}1\\ -1\end{bmatrix}\frac{EA\alpha\Delta T}{\sqrt{2}}$$

dvs

$$\begin{bmatrix}\delta_x\\ \delta_y\end{bmatrix}=\frac{\alpha\Delta T L}{3\sqrt{2}+4}\begin{bmatrix}4\\ -(4+2\sqrt{2})\end{bmatrix}$$

> 0 dvs nedåt

< 0 dvs åt vänster

Med δ_x δ_y kända fås nu (mha insättn.):

$$N_1=\frac{EA}{L}(\delta_x+\delta_y)=\ldots=-EA\alpha\Delta T\frac{2\sqrt{2}}{3\sqrt{2}+4}$$

$$N_2=\frac{EA}{L}\delta_x=\ldots=EA\alpha\Delta T\frac{4}{3\sqrt{2}+4}$$

$$N_3=\frac{EA}{L}\left(\frac{\delta_x-\delta_y}{2}-\alpha\Delta T L\right)=\ldots=-EA\alpha\Delta T\frac{2\sqrt{2}}{3\sqrt{2}+4}$$